THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — MATHS TWO-CHAPTER TEST SET C

Chapters 5 & 6: **Circles, Measuring Space: Perimeter and Area** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Draw neat figures wherever required.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. In a circle of radius 17 cm, a chord is 8 cm from the centre. The length of the chord is: 1

(a) 28 cm **(b)** 30 cm **(c)** 32 cm **(d)** 15 cm

2. In a circle of radius 10 cm, a chord is 8 cm from the centre. The length of the chord is: 1

(a) 10 cm **(b)** 12 cm **(c)** 14 cm **(d)** 6 cm

3. A chord of length 48 cm is at a distance of 7 cm from the centre of a circle. The radius of the circle is: 1

(a) 25 cm **(b)** 48 cm **(c)** 26 cm **(d)** 31 cm

4. A chord of length 16 cm is drawn in a circle of radius 17 cm. The distance of the chord from the centre is: 1

(a) 25 cm **(b)** 17 cm **(c)** 15 cm **(d)** 9 cm

5. An arc subtends an angle of 100° at the centre. The angle it subtends at a point on the remaining part of the circle is: 1

(a) 100° **(b)** 50° **(c)** 200° **(d)** 130°

6. Taking $\pi = \frac{22}{7}$, the circumference of a circle of radius 3.5 cm is: 1

(a) 44 cm **(b)** 22 cm **(c)** 11 cm **(d)** 38.5 cm

7. The perimeter of a circle is 132 cm. Taking $\pi = \frac{22}{7}$, its radius is: 1

(a) 10.5 cm **(b)** 42 cm **(c)** 22 cm **(d)** 21 cm

8. The length of an arc subtending 180° at the centre of a circle of radius 14 cm ($\pi = \frac{22}{7}$) is: 1

(a) 44 cm **(b)** 46 cm **(c)** 22 cm **(d)** 88 cm

9. By Heron's formula, the area of a triangle with sides 3 cm, 4 cm and 5 cm is: 1

(a) 6 cm^2 **(b)** 4 cm^2 **(c)** 18 cm^2 **(d)** 12 cm^2

10. By Heron's formula, the area of a triangle with sides 10 cm, 17 cm and 21 cm is: 1

(a) 90 cm^2 **(b)** 170 cm^2 **(c)** 78 cm^2 **(d)** 84 cm^2

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. ABCD is a cyclic quadrilateral with $\angle A = 3x^\circ$ and $\angle C = (2x + 30)^\circ$. Find x and $\angle A$. 2

12. An arc subtends an angle of 50° at a point on the circle. What angle does it subtend at the centre? 2

13. An isosceles triangle has equal sides 5 cm and base 6 cm. Find its area. 2

14. Find the area of a triangle with sides 7 cm, 15 cm and 20 cm. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Three friends stand at three points of a park that are not in a line. Where should a fountain be placed so that it is equally far from all three? Justify. 3

16. Baudhāyana's method: find the side of a square whose area equals that of a $16\ \text{cm} \times 9\ \text{cm}$ rectangle. 3

17. Find the centre and radius of the circle through $(0, 0)$, $(6, 0)$ and $(0, 8)$. 3

OR

Prove that equal chords of a circle subtend equal angles at the centre. 3

18. ABCD is a parallelogram of area 84 cm^2. Find the area of $\triangle ABD$ and justify. 3

OR

Find the perimeter of a sector of radius 21 cm and angle 60° ($\pi = \frac{22}{7}$). 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Prove that a cyclic parallelogram is a rectangle and that its diagonals are diameters of the circle. 5

OR

Case study: A ground is a circle of radius 13 m. Two parallel boundary ropes (chords) are 10 m and 24 m long. 5

(i) Find the distance of each from the centre.

(ii) How far apart are they if on opposite sides of the centre?

(iii) How far if on the same side?

(iv) Which rope is longer and which is nearer to the centre?

20. (i) List the integer-sided rectangles of perimeter 40 units and find their areas. 5

(ii) Which has the greatest area and which the least?

(iii) What do you notice about the shape of the rectangle of greatest area?

OR

(i) Prove that the two paths from P to Q — one a semicircle and the other made of n semicircles on adjacent parts of PQ — have equal length. 5

(ii) Hence find the length of each path if $PQ = 28$ cm and the smaller semicircles have diameters 7, 7, 14 cm. ($\pi = \frac{22}{7}$)

END OF QUESTION PAPER — Total: 40 Marks

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